

QURE Lab

Quantum Uncertainty-aware Research Environment · Research report · Qiskit Fall Fest 2026, Track 5 (Open Innovation), VIT Chennai

In one paragraph. We trained a 4-qubit Qiskit classifier on real lung-nodule data (597 LIDC-IDRI patients) and ran it under five simulated noise settings, with and without readout-error mitigation. The question is not whether quantum beats classical. It asks: when noise changes the classifier's answer, can we notice, and is mitigation worth what it costs? On validation, the quantum model (0.857 balanced accuracy) and logistic regression (0.810 at threshold 0.5, 0.845 val-tuned) are **not distinguishable**. At the highest noise, mitigation cut the score error from 0.141 to 0.041 but made run-to-run spread larger (0.011 $\rightarrow$ 0.019).

How to read this report

- **Simulated.** No real quantum hardware was used; the noise settings are hypothetical, not taken from a device.
- **Labels mean radiologist suspicion** (rating 1-2 = low, 4-5 = high), not confirmed cancer.
- **Validation numbers are optimistic** for the quantum model: its threshold was chosen on the same 90 cases.
- **Test results are pending.** Role 2 holds the 90 test labels; predictions were frozen and handed over.
- **Not a diagnostic tool.** Nothing here should be used to make decisions about a patient.

1 · Question

Near-term quantum computers are noisy. A quantum classifier's output is estimated from a finite number of measurements (shots), and gate and readout errors shift that estimate. For a decision task, the practical risk is a decision flip: noise pushes a case across the threshold. We study (a) how often that happens, (b) whether a rule that uses only what real hardware reports can flag the affected cases, and (c) what readout-error mitigation buys and costs.

2 · Data

Role 2 prepared the primary cohort from CIRDataset (Zenodo record 6762573, CC BY 4.0 (Zenodo record); cite original LIDC-IDRI as well), which supplies LIDC-IDRI CT patches with expert nodule masks. One canonical nodule per patient was kept, giving 597 patients (295 high, 302 low suspicion). Rating-3 cases were excluded.

Split (by patient)	Patients	Labels
Train	417	used to fit the model and scaling
Validation	90	used to pick restart and threshold; reported here
Test	90	held by Role 2 until predictions were frozen

Four image/mask features were computed per nodule: `log_mask_voxels`, `mean_stored_intensity`, `std_stored_intensity`, `bbox_fill_fraction`. Each was standardised with train-split statistics and mapped to a rotation angle $\theta = \pi \cdot \text{sigmoid}(z)$, so θ

lies in $(0, \pi)$.

3 • Method

Circuit

Each angle θ_i rotates qubit i (RY). Two layers follow, each with a trainable RY on every qubit and a chain of CX gates (0-1, 1-2, 2-3), then a final trainable RY layer: 12 trainable weights. All four qubits are measured; the **score** is the share of shots with an odd number of 1s. The decision is “high suspicion” when $\text{score} \geq 0.39$. The score is a measurement result, not a probability of disease.

Training

Class-balanced cross-entropy on noiseless scores of the train split, COBYLA, 3 random restarts. The restart with the lowest validation loss was kept, the threshold maximised validation balanced accuracy, and all weights were then frozen.

Noise and mitigation

Setting	Description	1-qubit depol.	2-qubit depol.	P(1 0)	P(0 1)
N0	noiseless	0.0	0.0	0.0	0.0
N1	readout only	0.0	0.0	0.02	0.04
N2	gate only	0.001	0.01	0.0	0.0
N3	gate + readout (moderate)	0.001	0.01	0.02	0.04
N4	gate + readout (high)	0.003	0.03	0.05	0.08

Every case was run 5 times per setting with 2000 shots (Qiskit Aer, basis rz/sx/x/cx, no transpiler optimisation). **Readout mitigation** measures how often each qubit's 0 and 1 are misread (4000 calibration shots) and inverts that per-qubit error. It does not touch gate noise. Mitigated values outside $[0, 1]$ are kept, never clipped. A raw run with the same total of 6000 shots shows what the calibration shots would buy if spent on plain sampling.

Instability rule U (declared before the test run)

Flag a case when the raw and mitigated averages give different decisions, or when the mitigated average is within two standard errors of the threshold. It uses only quantities a real device would report. It is a prompt for review, not a trust score.

4 • Results

Prediction quality (validation, 90 cases, noiseless)

Model	Thr.	Balanced acc. (95% CI)	AUC	Sensitivity	Specificity
4-qubit classifier (CX chain)	0.39	0.857 (0.780–0.928)	0.911	0.909	0.804
4-qubit, no CX (ablation)	0.49	0.846 (0.774–0.913)	0.893	0.909	0.783
Logistic regression (Role 2)	0.5	0.810 (0.730–0.890)	0.897	0.773	0.848
Logistic regression, val-tuned	0.34	0.845 (0.772–0.916)	0.897	0.886	0.804

Paired bootstrap of the difference (quantum minus val-tuned logistic regression): +0.011, 95% CI -0.022 to +0.050. The interval includes zero, so the two models cannot be told apart on 90 cases. We make no claim of quantum advantage.

Noise, mitigation and cost

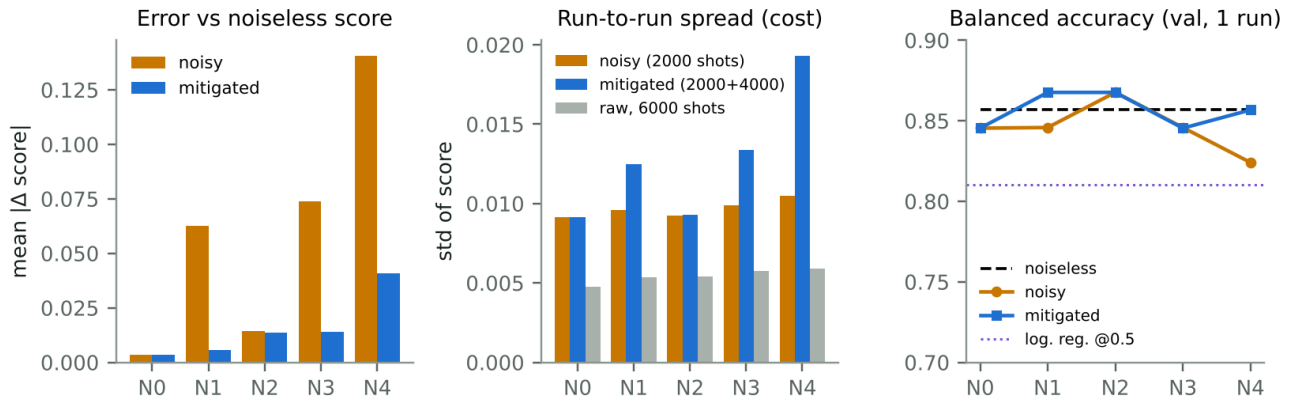


Figure 1. Validation split. Left: mitigation removes most of the readout bias (N1, N3, N4) but not gate noise (N2). Middle: the price is a larger spread between runs; a raw run with the same shot budget is steadier still. Right: single-run balanced accuracy stays within a few points of the noiseless reference.

Setting	$ \Delta $ noisy	$ \Delta $ mitigated	Flip rate noisy	Flip rate mitigated	Std noisy	Std mitigated	Std raw 6000
N0	0.004	0.004	0.9%	0.9%	0.009	0.009	0.005
N1	0.057	0.005	4.0%	1.1%	0.010	0.013	0.005
N2	0.014	0.013	1.3%	1.1%	0.009	0.009	0.005
N3	0.069	0.013	5.6%	1.8%	0.009	0.013	0.006
N4	0.131	0.038	16.9%	3.3%	0.010	0.020	0.006

Table: test split (label-free reliability, 90 cases $\times$ 5 repeats). Flip rate = share of repeats whose decision differs from the noiseless one.

Instability rule U

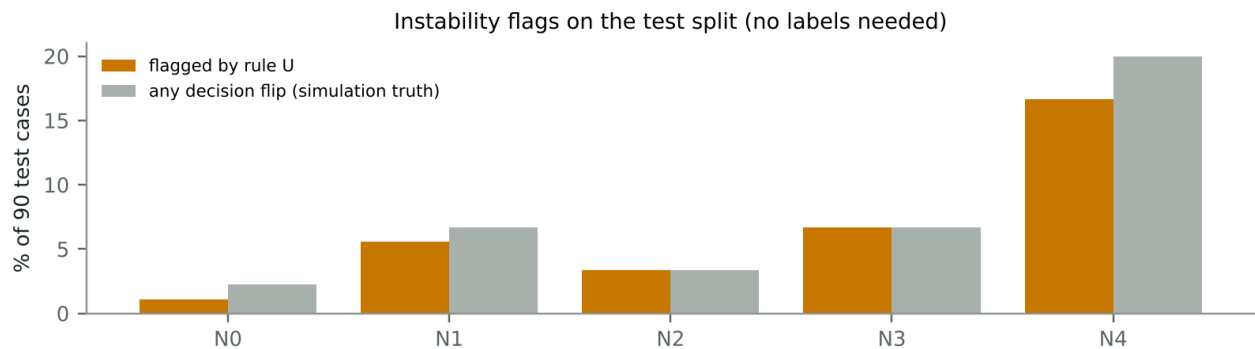


Figure 2. At N4, rule U flagged 16.7% of test cases; every flagged case really had a flipping run (precision 100.0%), and it caught 83.3% of the 18 cases that flipped. Counts at lower noise are small, so treat them as indicative.

Replication cohort

An earlier cohort built by Role 1 from LIDC reader contours (different features, 183 test nodules with labels) gave quantum 0.857 (AUC 0.882) vs logistic regression 0.852 (AUC 0.919) on test, and removing the CX gates dropped the quantum model to 0.607. On the primary cohort the ablation barely matters, so we do not generalise about entanglement either way.

5 • What we can and cannot say

Supported

- A 4-qubit simulated classifier reaches validation accuracy comparable to logistic regression on the same four inputs.
- Readout mitigation removes most readout-induced score error in this simulation, at the cost of 3× the shots and a larger run-to-run spread.
- A hardware-observable rule can flag many cases whose decision is unstable under noise.

Not supported

- Any quantum advantage.
- Clinical validity, diagnosis or use on patients.
- Behaviour on real quantum hardware (the noise settings are hypothetical).
- Test-split accuracy on the primary cohort (pending Role 2's scoring).

Known limits (from the data card)

- Selected clear-rating cases; not representative of all nodules.
- Expert-provided segmentations required, not automated diagnosis from arbitrary CT.
- Stored intensities are preprocessed, not raw HU.
- Voxel-grid size is not physical volume and can depend on resampling.
- Prediction scores are not established calibrated clinical probabilities.
- One validation split; confidence interval is conditional on this split.
- 8 alternate-scan identifiers excluded rather than guessing label mappings.

6 • Using the dashboard

- **Start** gives the three most common tasks. Switch between **Simple** and **Detailed** view in the top bar.
- **Cases**: pick a nodule and a noise setting to see its noiseless, noisy and mitigated scores.
- **Accuracy lab**: move the threshold or noise setting and watch the confusion matrix and accuracy update (validation only).
- **Ask QURE**: the assistant answers common questions from this report's numbers; it does not give medical advice.

Glossary

Term	Meaning here
Balanced accuracy	Average of sensitivity and specificity; fair when classes are uneven.
Sensitivity / specificity	Share of high-suspicion cases called high / low-suspicion cases called low.
AUC	How well scores rank high above low cases, over all thresholds (0.5 = chance).
Shot	One run-and-measure of the circuit. Scores are averages over shots.
Readout error	A qubit measured as the wrong value, e.g. 1 read as 0.
Mitigation	Post-processing that estimates the error-free result; it does not repair the hardware.
Decision flip	A run whose decision differs from the noiseless reference.
Bloch vector	Arrow describing one qubit's state; shorter than 1 when qubits are entangled.

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